

Kendrick Robinson Portfolio

Machine Learning Engineer specializing in classification algorithms, data analysis, and model evaluation techniques

by Kendrick Robinson

Introduction to Machine Learning ITAI1371



Learning Journey

From fundamentals to advanced ML concepts

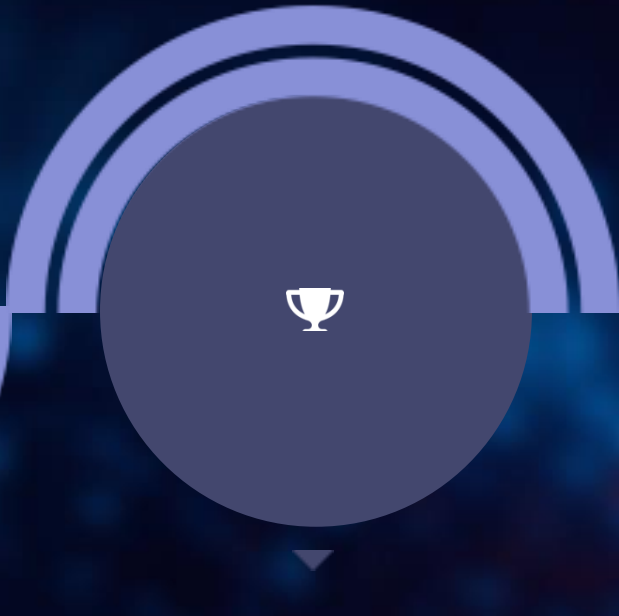
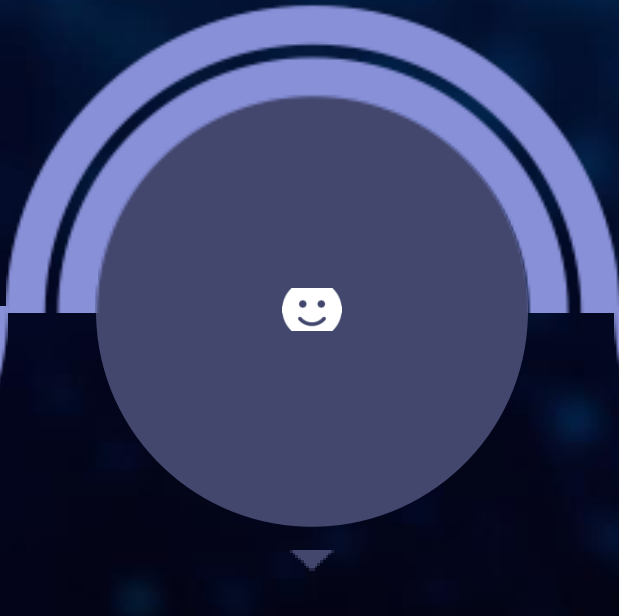
Module 2: ML Tools Setup

Python, Jupyter, pandas, NumPy, matplotlib, scikit-learn



Midterm Exam: Classification Models

Logistic Regression, Decision Trees, Cross-validation

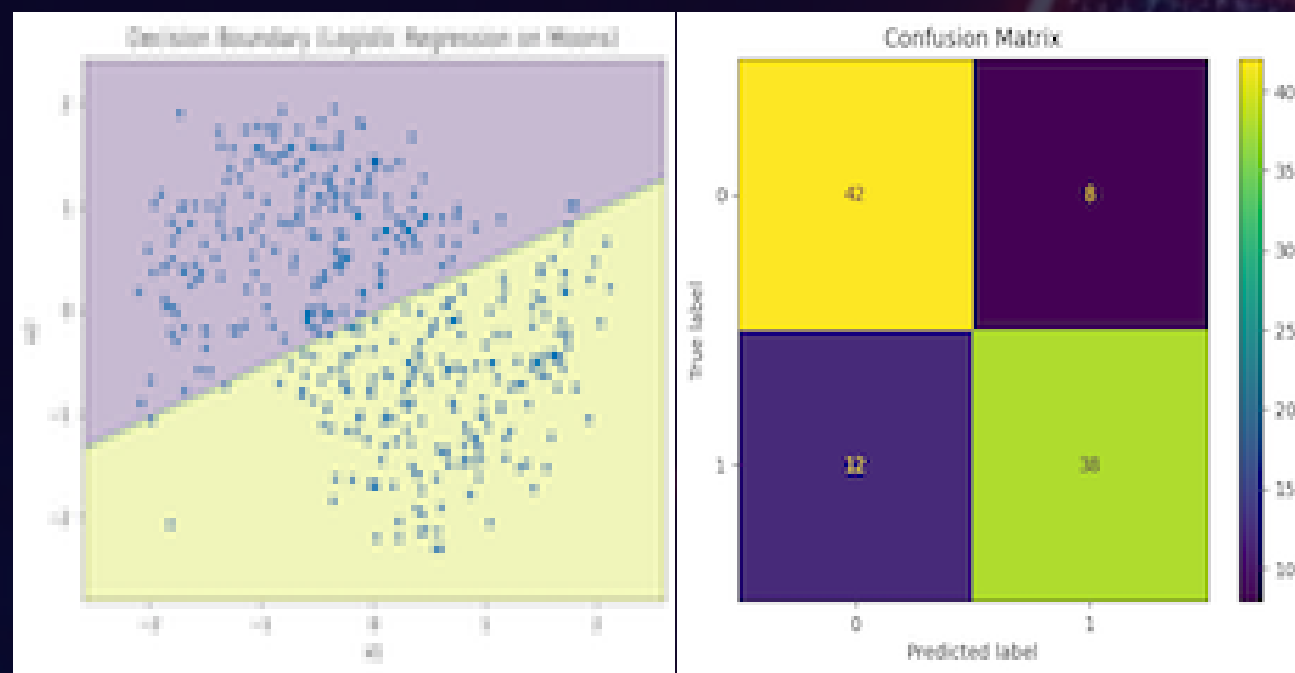


Module 3: ML Workflow & Types

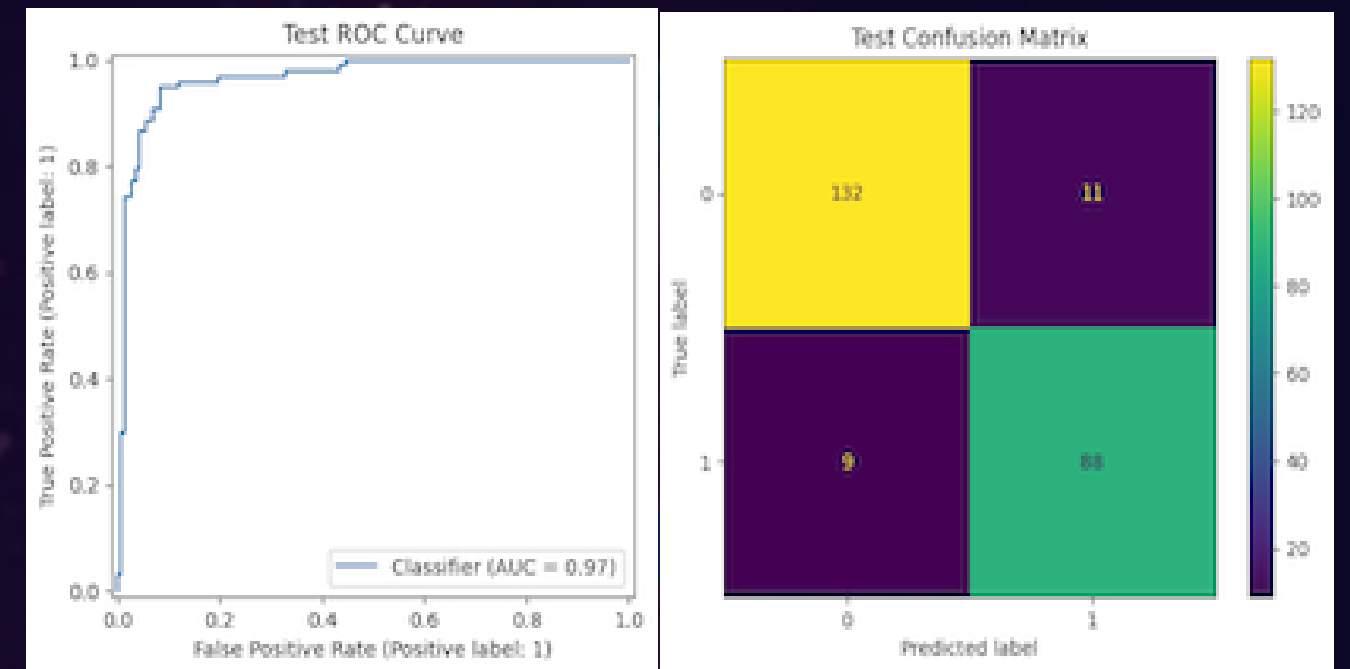
Supervised/Unsupervised learning, Wine classification project

Final Project: Advanced Pipeline

GridSearchCV, Random Forest, Model evaluation with confusion matrix and ROC curves



Featured Projects



Midterm Exam – Developing Notebook (Introduction to ML)

Implemented binary classification on Moons dataset using Logistic Regression. Analyzed decision boundaries on non-linearly separable data, achieving 80% accuracy with comprehensive model evaluation.

Technologies: scikit-learn, matplotlib, confusion matrix analysis

Decision boundary visualization on synthetic dataset

Final ML Project: Binary Classification Pipeline

Implemented end-to-end ML pipeline with preprocessing, hyperparameter tuning via GridSearchCV, achieving 91.67% accuracy on test data.

Technologies: Python, scikit-learn, Random Forest, Logistic Regression

ROC curve visualization (AUC = 0.97)

Skills & Competencies

100%

Python Proficiency

Technical Skills

- Proficient in essential machine learning tools and frameworks for data science projects
- Python, Jupyter Notebooks, Google Colab environment setup
- Data manipulation with pandas, NumPy for numerical computing
- Visualization using matplotlib, seaborn for insights generation

95%

Data Analysis

ML Competencies

- Hands-on experience with supervised learning algorithms and model evaluation techniques
- Classification models: Logistic Regression, Decision Trees, Random Forest
- Model validation: Cross-validation, GridSearchCV hyperparameter tuning
- Performance metrics: Confusion matrices, ROC curves, AUC scoring

85%

ML Algorithms

90%

Model Evaluation

Soft Skills

- Strong analytical mindset with ability to communicate complex technical concepts clearly
- Problem-solving approach to data science challenges systematically
- Documentation and code organization for reproducible research
- Critical thinking for model interpretation and bias detection

Reflection & Future Goals

Continuous learning in machine learning



Learning Reflection

Gained solid foundation in ML fundamentals through hands-on projects. Developed expertise in classification algorithms, model evaluation, and ethical AI considerations. Understand the importance of data preprocessing and validation techniques.



Next Steps

Learn TensorFlow/PyTorch for deep learning, Deploy ML model as REST API, Build comprehensive portfolio website, Explore advanced topics: NLP, Computer Vision

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